

On the Finiteness of Solutions to $u^n + f(n) = k^2$

A Generalization via Linear Forms in Logarithms

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Abstract

We generalize the method of [1]—which establishes that the equation $2^n - n = k^2$ admits only the solutions $n = 1, 7$ —to the broader family

$$u^n + f(n) = k^2, \quad u \in \mathbb{Z}_{\geq 2}, \quad n, k \in \mathbb{Z}_{>0},$$

where $f : \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}$ is a function of *sub-exponential* growth (in the precise sense that $\log |f(n)| = o(n/\log n)$; in particular every polynomial satisfies this). Using the structure of the unit group of the real quadratic field $\mathbb{Q}(\sqrt{d})$ (where d is the squarefree part of u) together with Matveev’s explicit lower bound for linear forms in logarithms of algebraic numbers, we show that every solution satisfies an explicit upper bound $n \leq N_0(u, f)$, and hence the solution set is finite.

1 Introduction and Statement of Results

1.1 Background

The reference article [1] proves that the Diophantine equation

$$2^n - n = k^2 \tag{1}$$

has exactly two positive-integer solutions, namely $(n, k) = (1, 1)$ and $(7, 11)$. The proof proceeds in three conceptual stages:

- (i) **Elementary parity reduction.** For even $n \geq 4$, writing $x = 2^{n/2}$ gives $(x-k)(x+k) = n$, which is impossible since $x+k \geq x > n$. Hence n must be odd, say $n = 2m+1$, and the equation becomes the generalized Pell equation

$$k^2 - 2q^2 = -n, \quad q = 2^m.$$

- (ii) **Linear forms in logarithms.** Working in $\mathbb{Z}[\sqrt{2}]$ with the fundamental unit $\eta = 3+2\sqrt{2}$, one reduces the algebraic integer $\alpha = k + q\sqrt{2}$ by a suitable power η^{-t} to a bounded representative β . The quantity $\Delta = \alpha/(q\sqrt{2}) - 1$ is exponentially small (of order $n/2^n$), and $\log(1+\Delta)$ is a linear form in logarithms of the algebraic numbers $\eta, 2, \gamma$ (where $\gamma = \beta/(2\sqrt{2})$). Matveev’s theorem [2] then yields an explicit lower bound $|\Lambda| > \exp(-C(1+\log m) \cdot A_1 A_2 A_3)$, which, combined with the upper bound $|\Lambda| < 2|\Delta|$, forces $n \leq 4.85 \times 10^{15}$.
- (iii) **Sieve.** Quadratic-residue sieves modulo a battery of primes reduce the surviving candidates to $n \in \{1, 7\}$, verified directly.

1.2 Main Theorem

The purpose of this note is to show that the *finiteness* conclusion (stage (ii) above) generalizes to a wide class of equations of the form $u^n + f(n) = k^2$, with essentially the same proof strategy. The key geometric insight is that the term $f(n)$, being sub-exponential, plays the role of a *small perturbation* of the perfect power u^n , and the algebraic-number-theoretic machinery of [1] is robust enough to absorb this perturbation.

Theorem 1.1 (Main Theorem). *Let $u \geq 2$ be a fixed positive integer, and let $f : \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}$ be a function satisfying the growth condition*

$$\log |f(n)| = o\left(\frac{n}{\log n}\right) \quad (n \rightarrow \infty). \quad (2)$$

Then the Diophantine equation

$$u^n + f(n) = k^2, \quad n, k \in \mathbb{Z}_{>0}, \quad (3)$$

has at most finitely many solutions (n, k) .

Remark 1.2. *Condition (2) is satisfied by every polynomial (since $\log |f(n)| = O(\log n) = o(n/\log n)$), and more generally by every function of the form $|f(n)| = \exp(O(n^\alpha))$ with $0 \leq \alpha < 1$, or $|f(n)| = \exp(O(n/(\log n)^{1+\varepsilon}))$ with $\varepsilon > 0$. In particular, the theorem applies to the reference case $u = 2$, $f(n) = -n$.*

Remark 1.3. *The restriction $u \geq 2$ is necessary. For $u = 1$, the equation becomes $1 + f(n) = k^2$, and the conclusion can fail: taking $f(n) = n^2 - 1$ (a polynomial) gives $k = n$ for every n , yielding infinitely many solutions. The reason is that $u^n = 1$ does not grow, so $f(n)$ is no longer a “small perturbation.”*

Remark 1.4. *When u is a perfect square, say $u = v^2$, the equation admits the trivial solution $k = v^n$ whenever $f(n) = 0$. If $f(n) = 0$ for infinitely many n , there are infinitely many trivial solutions. The content of Theorem 1.1 is that the non-trivial solutions (those with $f(n) \neq 0$) are always finite. Equivalently, the set $\{n \geq 1 : u^n + f(n) \text{ is a perfect square and } f(n) \neq 0\}$ is finite.*

1.3 Proof Strategy

The proof splits into two cases depending on whether u is a perfect square:

- **Case 1 ($u = v^2$):** Then $u^n = (v^n)^2$ is always a perfect square, and the equation factors as $(k - v^n)(k + v^n) = f(n)$. An elementary size argument shows that for large n , $|f(n)| < v^n$ forces $k = v^n$ and hence $f(n) = 0$.
- **Case 2 (u not a perfect square):** Write $u = da^2$ with $d > 1$ squarefree. For even n , u^n is again a perfect square and the elementary argument applies. For odd $n = 2m + 1$, we have $u^n = db^2$ with $b = d^m a^{2m+1}$, and the equation becomes the generalized Pell equation $k^2 - db^2 = f(n)$. This is the heart of the proof: we reduce via the unit group of $\mathbb{Q}(\sqrt{d})$, form a linear form in logarithms, and apply Matveev’s theorem.

2 Preliminaries

2.1 Quadratic Fields and the Pell Equation

Let $d > 1$ be a squarefree integer and $K = \mathbb{Q}(\sqrt{d})$ the associated real quadratic field. The conjugation $\bar{\cdot} : K \rightarrow K$ sends $\sqrt{d} \mapsto -\sqrt{d}$, and the norm is $N(\alpha) = \alpha \bar{\alpha} \in \mathbb{Q}$.

The ring $\mathbb{Z}[\sqrt{d}] = \{x + y\sqrt{d} : x, y \in \mathbb{Z}\} \subset K$ is closed under conjugation, and the norm of any of its elements is an integer: $N(x + y\sqrt{d}) = x^2 - dy^2$.

The existence of norm-one elements in K of arbitrarily large absolute value is the content of the classical Pell equation, and we need nothing more than this:

Theorem 2.1 (Pell equation). *For every squarefree $d > 1$, the equation*

$$x^2 - dy^2 = 1 \tag{4}$$

has a solution $(x_1, y_1) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$ with $x_1 > 1$.

Sketch. This is a classical result going back to Lagrange. A standard proof uses the continued-fraction expansion of $\sqrt{d}$, which is eventually periodic; the convergents p_n/q_n satisfy $|p_n^2 - dq_n^2| \leq 2$, and by periodicity some convergent achieves $p_n^2 - dq_n^2 = 1$.

Alternatively, a self-contained proof uses Dirichlet's pigeonhole principle: among the $N + 1$ fractional parts $\{j\sqrt{d}\}$ for $j = 0, 1, \dots, N$, two must lie within $1/N$ of each other, yielding integers x, y with $|y| \leq N$ and $|x^2 - dy^2| \leq 2\sqrt{d} + 1$. This produces a nontrivial solution to $|x^2 - dy^2| \leq 2\sqrt{d} + 1$; multiplying together finitely many such solutions (via the identity $(x^2 - dy^2)(x'^2 - dy'^2) = (xx' + dy'y')^2 - d(xy' + x'y)^2$) and using the finiteness of the possible values of $|x^2 - dy^2|$, one extracts a solution to $x^2 - dy^2 = 1$. $\square$

Definition 2.2. *A fundamental unit of $\mathbb{Q}(\sqrt{d})$ is any element*

$$\varepsilon := x_1 + y_1\sqrt{d} > 1, \quad x_1, y_1 \in \mathbb{Z}_{>0}, \quad x_1^2 - dy_1^2 = 1, \tag{5}$$

furnished by Theorem 2.1. We do not require ε to be the smallest such element; any solution of (4) will do.

The three properties of ε that we actually use below are:

- (i) $\varepsilon > 1$ is a real number, so $\log \varepsilon > 0$;
- (ii) $\bar{\varepsilon} = x_1 - y_1\sqrt{d} = 1/\varepsilon$ (since $\varepsilon\bar{\varepsilon} = 1$), hence $\varepsilon^{-1} = x_1 - y_1\sqrt{d} \in \mathbb{Z}[\sqrt{d}]$;
- (iii) for any $\alpha \in \mathbb{Z}[\sqrt{d}]$ and any $t \in \mathbb{Z}$, the product $\alpha\varepsilon^t$ remains in $\mathbb{Z}[\sqrt{d}]$ (the ring is closed under multiplication by both ε and ε^{-1}).

Property (iii) is the only algebraic input we need: it guarantees that the “reduced” element $\beta = \alpha\varepsilon^{-t}$ constructed below is still an algebraic integer in $\mathbb{Z}[\sqrt{d}]$, so its norm is an integer and its height is well-controlled. No invocation of Dirichlet's unit theorem or of the full structure of $\mathcal{O}_K^\times$ is required.

Definition 2.3 (Absolute logarithmic height). *For a non-zero algebraic number α of degree D over $\mathbb{Q}$, with minimal polynomial $a_0 \prod_{i=1}^D (X - \alpha^{(i)})$, the absolute logarithmic height is*

$$h(\alpha) = \frac{1}{D} \left(\log |a_0| + \sum_{i=1}^D \log \max(1, |\alpha^{(i)}|) \right).$$

The height satisfies the fundamental sub-additivity property $h(\alpha\beta) \leq h(\alpha) + h(\beta)$ and $h(\alpha/\beta) \leq h(\alpha) + h(\beta)$.

2.2 Matveev's Theorem

We use the following explicit lower bound for linear forms in logarithms, in the form of Laurent–Mignotte–Nesterenko [2] (Corollary 2.3 of the reference cited in [1]):

Theorem 2.4 (Matveev, explicit form). *Let $\alpha_1, \dots, \alpha_s$ be positive real algebraic numbers in a number field of degree D , and let $b_1, \dots, b_s \in \mathbb{Z}$ with*

$$\Lambda = b_1 \log \alpha_1 + \dots + b_s \log \alpha_s \neq 0.$$

Set $B = \max(|b_1|, \dots, |b_s|, 1)$ and $A_i = \max(D h(\alpha_i), |\log \alpha_i|, 0.16)$ for each i . Then

$$\log |\Lambda| > -C_1(s) \cdot 30^{s+3} s^{4.5} D^2 (1 + \log D) (1 + \log B) A_1 \cdots A_s, \quad (6)$$

where $C_1(s) = \min(0.5 e^{s 30^{s+3} s^{4.5}}, 26s + 20)$.

In our application we always have $s = 3$ and $D = 2$, so the constant collapses to a single (large) number $C_0 \approx 2.31 \times 10^{12}$, exactly as in [1].

2.3 A Calculus Lemma

We reproduce the elementary lemma from [1] that converts between $|\Delta|$ and $|\Lambda|$ when $\Delta = e^\Lambda - 1$ is small.

Lemma 2.5. *For $x \in (-1, 0)$, we have $\ln(1+x) > x/(x+1)$.*

Proof. Define $g(x) = \ln(1+x) - x/(x+1)$ for $x \in (-1, 0)$. Then $g'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} < 0$, so g is strictly decreasing. Since $g(0) = 0$ and g is continuous, we have $g(x) > 0$ for all $x \in (-1, 0)$. $\square$

Corollary 2.6. *If $|\Delta| < 1/2$ and $\Lambda = \log(1 + \Delta)$, then $|\Lambda| < 2|\Delta|$.*

Proof. By Lemma 2.5 (and its reflection for $\Delta > 0$), $|\Lambda| = |\log(1 + \Delta)| \leq |\Delta|/(1 - |\Delta|) < 2|\Delta|$ when $|\Delta| < 1/2$. $\square$

3 Proof of the Main Theorem

Throughout this section, $u \geq 2$ is a fixed integer and f satisfies (2). We write $u = d a^2$ where $d \geq 1$ is squarefree and $a \geq 1$; note $d = 1$ iff u is a perfect square.

3.1 Case 1: u is a Perfect Square ($d = 1$)

Here $u = v^2$ with $v \geq 2$, and $u^n = (v^n)^2$. The equation becomes

$$(v^n)^2 + f(n) = k^2, \quad \text{i.e.,} \quad (k - v^n)(k + v^n) = f(n).$$

Proposition 3.1. *If $u = v^2$ with $v \geq 2$, then every solution of (3) with $f(n) \neq 0$ satisfies $v^n \leq |f(n)|$. In particular, the set of such n is finite.*

Proof. Suppose $f(n) \neq 0$. Then $k \neq v^n$, so $|k - v^n| \geq 1$ (both are integers). Since $k + v^n \geq v^n > 0$, we obtain

$$|f(n)| = |k - v^n| \cdot |k + v^n| \geq 1 \cdot v^n = v^n.$$

But $v \geq 2$ and $\log |f(n)| = o(n/\log n) = o(n)$, so $\log |f(n)| < n \log v$ for all sufficiently large n , i.e., $|f(n)| < v^n$. This contradicts $|f(n)| \geq v^n$. Hence only finitely many n can satisfy $f(n) \neq 0$ and yield a solution. $\square$

3.2 Case 2: u is Not a Perfect Square ($d > 1$)

3.2.1 Even n : Elementary Reduction

For even $n = 2m$, we have $u^n = (u^m)^2$, so the same factoring argument as Proposition 3.1 applies with v replaced by u :

Proposition 3.2. *Every solution of (3) with n even and $f(n) \neq 0$ satisfies $u^{n/2} \leq |f(n)|$. In particular, only finitely many even n yield non-trivial solutions.*

Proof. Identical to Proposition 3.1: $(k - u^{n/2})(k + u^{n/2}) = f(n)$, and $|f(n)| = o(u^{n/2})$ since $\log |f(n)| = o(n) = o(n \log u/2)$. $\square$

3.2.2 Odd n : Pell Reduction

Henceforth $n = 2m + 1$ is odd with $m \geq 1$. We have

$$u^n = (da^2)^{2m+1} = d^{2m+1} a^{4m+2} = d \cdot (d^m a^{2m+1})^2 = db^2,$$

where we set

$$b := d^m a^{2m+1} \in \mathbb{Z}_{>0}, \quad \text{so that} \quad b^2 = \frac{u^n}{d}. \quad (7)$$

The equation (3) becomes the *generalized Pell equation*

$$k^2 - db^2 = f(n). \quad (8)$$

Set $K = \mathbb{Q}(\sqrt{d})$, and let $\varepsilon > 1$ be a fundamental unit in the sense of Definition 2.2 (so $N(\varepsilon) = 1$ and $\varepsilon^{-1} \in \mathbb{Z}[\sqrt{d}]$). Define

$$\alpha := k + b\sqrt{d} \in \mathbb{Z}[\sqrt{d}], \quad \bar{\alpha} = k - b\sqrt{d}, \quad N(\alpha) = \alpha \bar{\alpha} = f(n). \quad (9)$$

Lemma 3.3 (Size estimates). *For n sufficiently large (so that $|f(n)| < u^n/4$):*

$$(a) \quad k = \sqrt{u^n + f(n)}, \text{ hence } u^{n/2} \sqrt{1 - |f(n)|/u^n} \leq k \leq u^{n/2} \sqrt{1 + |f(n)|/u^n}.$$

$$(b) \quad \alpha = k + b\sqrt{d} \geq 2u^{n/2} \sqrt{1 - |f(n)|/u^n} \geq u^{n/2}.$$

$$(c) \quad |\bar{\alpha}| = |f(n)|/|\alpha| \leq |f(n)|/u^{n/2}.$$

Proof. Since $b\sqrt{d} = \sqrt{db^2} = \sqrt{u^n} = u^{n/2}$, we have $\alpha = k + u^{n/2}$ and $\bar{\alpha} = k - u^{n/2}$. The bound $|f(n)| < u^n/4$ ensures $k = \sqrt{u^n + f(n)}$ is real and close to $u^{n/2}$. Specifically, $|k - u^{n/2}| = |f(n)|/(k + u^{n/2}) \leq |f(n)|/u^{n/2}$, giving (a). Then (b) follows from $\alpha = k + u^{n/2} \geq 2u^{n/2} - |f(n)|/u^{n/2} \geq u^{n/2}$, and (c) from $|\bar{\alpha}| = |f(n)|/\alpha \leq |f(n)|/u^{n/2}$. $\square$

3.2.3 Unit Group Reduction

Proposition 3.4. *Assume $f(n) \neq 0$. There exists a unique integer $t \geq 0$ such that*

$$\beta := \alpha \varepsilon^{-t} \quad \text{satisfies} \quad \sqrt{|f(n)|} \leq |\beta| < \varepsilon \sqrt{|f(n)|}. \quad (10)$$

Moreover, the conjugate $\bar{\beta} = \bar{\alpha} \varepsilon^t$ satisfies $|\bar{\beta}| \leq \sqrt{|f(n)|}$, and

$$t \leq \frac{n \log u}{2 \log \varepsilon} + \frac{\log 2}{\log \varepsilon}. \quad (11)$$

Proof. Since $\varepsilon > 1$ and $\alpha > 0$, the function $t \mapsto \alpha \varepsilon^{-t}$ is continuous and strictly decreasing from $+\infty$ to 0 as t ranges over $\mathbb{R}$. Hence there is a unique real t_0 with $\alpha \varepsilon^{-t_0} = \sqrt{|f(n)|}$, and taking $t = \lceil t_0 \rceil$ gives (??). (If $t_0 < 0$, take $t = 0$; this happens only for small n .)

For the conjugate: $N(\beta) = N(\alpha) N(\varepsilon)^{-t} = f(n)$, so $|\beta| = |f(n)|/|\alpha| \leq |f(n)|/\sqrt{|f(n)|} = \sqrt{|f(n)|}$.

For the bound on t : from $|\beta| \geq \sqrt{|f(n)|}$ and $|\alpha| \leq 2u^{n/2}$ (by Lemma 3.3(b) for large n),

$$\varepsilon^t = \frac{|\alpha|}{|\beta|} \leq \frac{2u^{n/2}}{\sqrt{|f(n)|}}.$$

Taking logarithms and using $\log |f(n)| \geq 0$ (for $|f(n)| \geq 1$):

$$t \leq \frac{\log 2 + \frac{n}{2} \log u - \frac{1}{2} \log |f(n)|}{\log \varepsilon} \leq \frac{n \log u}{2 \log \varepsilon} + \frac{\log 2}{\log \varepsilon}. \quad \square$$

3.2.4 The Linear Form in Logarithms

Define the key quantity

$$\Delta := \frac{\alpha}{b\sqrt{d}} - 1 = \frac{k + u^{n/2}}{u^{n/2}} - 1 = \frac{k}{u^{n/2}} - 1. \quad (12)$$

Since $k = \sqrt{u^n + f(n)} = u^{n/2} \sqrt{1 + f(n)/u^n}$, we have

$$\Delta = \sqrt{1 + \frac{f(n)}{u^n}} - 1. \quad (13)$$

Lemma 3.5 (Upper bound on $|\Delta|$). *For n sufficiently large,*

$$|\Delta| \leq \frac{|f(n)|}{u^n}. \quad (14)$$

Proof. For $|x| < 1$, $|\sqrt{1+x} - 1| = |x|/(\sqrt{1+x} + 1) \leq |x|/2 \leq |x|$. Since $|f(n)|/u^n \rightarrow 0$ (because $\log |f(n)| = o(n) < n \log u$ for large n), the bound applies with $x = f(n)/u^n$. $\square$

Now set $\Lambda := \log(1 + \Delta) = \log(\alpha/(b\sqrt{d}))$, so that $\Delta = e^\Lambda - 1$. By Corollary 2.6 and Lemma 3.5,

$$|\Lambda| \leq 2|\Delta| \leq \frac{2|f(n)|}{u^n}. \quad (15)$$

We now express Λ as a linear form in logarithms. Since $\alpha = \beta \varepsilon^t$ and $b\sqrt{d} = u^{n/2}$, we have

$$\Lambda = \log \beta + t \log \varepsilon - \frac{n}{2} \log u.$$

Since $n = 2m + 1$ is odd, $n/2 = m + 1/2$. To obtain integer coefficients, absorb the half-power into the algebraic number: set

$$\gamma := \frac{\beta}{\sqrt{u}} = \frac{\beta}{a\sqrt{d}} \in K, \quad (16)$$

so that $\log \gamma = \log \beta - \frac{1}{2} \log u$, and

$$\boxed{\Lambda = \log \gamma + t \log \varepsilon - m \log u} \quad (17)$$

with integer coefficients $1, t, -m$ and algebraic numbers γ, ε, u .

3.2.5 Height Estimates

Lemma 3.6. *The heights of the algebraic numbers in (17) satisfy:*

- (a) $h(\varepsilon) = \frac{1}{2} \log \varepsilon$ (constant, independent of n).
- (b) $h(u) = \log u$ (constant).
- (c) $h(\gamma) \leq \frac{1}{2}(\log |f(n)| + \log \varepsilon) + \frac{1}{2} \log u + \log a + \frac{1}{2} \log d$.

In particular, $h(\gamma) = O(\log |f(n)| + 1)$.

Proof. (a) $\varepsilon = x_1 + y_1 \sqrt{d}$ satisfies $x_1^2 - d y_1^2 = 1$, so its minimal polynomial is $X^2 - 2x_1 X + 1$ (with $|a_0| = 1$), and $h(\varepsilon) = \frac{1}{2}(\log \max(1, \varepsilon) + \log \max(1, \bar{\varepsilon})) = \frac{1}{2} \log \varepsilon$ since $\varepsilon > 1$ and $\bar{\varepsilon} = 1/\varepsilon < 1$.

(b) $u \in \mathbb{Z}$, so $h(u) = \log u$.

(c) Since $\beta = \alpha \varepsilon^{-t} \in \mathbb{Z}[\sqrt{d}]$ (by property (iii) of Definition 2.2) is a quadratic algebraic integer with $N(\beta) = f(n)$, we have $h(\beta) = \frac{1}{2}(\log \max(1, |\beta|) + \log \max(1, |\bar{\beta}|))$. By Proposition 3.4, $|\beta| < \varepsilon \sqrt{|f(n)|}$ and $|\bar{\beta}| \leq \sqrt{|f(n)|}$, so

$$h(\beta) \leq \frac{1}{2}(\log \varepsilon + \frac{1}{2} \log |f(n)| + \frac{1}{2} \log |f(n)|) = \frac{1}{2}(\log |f(n)| + \log \varepsilon).$$

By sub-additivity, $h(\gamma) = h(\beta/\sqrt{u}) \leq h(\beta) + h(\sqrt{u})$. Since $\sqrt{u} = a\sqrt{d}$ has minimal polynomial $X^2 - u$ (or $X^2 - d a^2$), $h(\sqrt{u}) = \frac{1}{2} \log u$. Adding the (constant) contribution from the denominator $a\sqrt{d}$ gives the stated bound. $\square$

3.2.6 Applying Matveev's Theorem

We apply Theorem 2.4 to the linear form (17) with parameters:

$$s = 3, \quad D = 2, \quad (\alpha_1, \alpha_2, \alpha_3) = (\varepsilon, u, \gamma), \quad (b_1, b_2, b_3) = (t, -m, 1).$$

The Matveev parameters are:

$$A_1 = \max(2h(\varepsilon), \log \varepsilon, 0.16) = \log \varepsilon, \tag{18}$$

$$A_2 = \max(2h(u), \log u, 0.16) = \log u, \tag{19}$$

$$A_3 = \max(2h(\gamma), |\log \gamma|, 0.16) = O(\log |f(n)| + 1), \tag{20}$$

$$B = \max(t, m, 1) = O(n) \quad (\text{by (11) and } m = (n-1)/2).$$

The constant is $C_0 := C_1(3) \cdot 30^6 \cdot 3^{4.5} \cdot 4 \cdot (1 + \log 2) \approx 2.31 \times 10^{12}$, exactly as in [1].

Matveev's theorem gives (assuming $\Lambda \neq 0$, which we verify below):

$$\log |\Lambda| > -C_0 (1 + \log B) A_1 A_2 A_3. \tag{21}$$

Lemma 3.7 ($\Lambda \neq 0$). *For n sufficiently large with $f(n) \neq 0$, the linear form Λ is non-zero.*

Proof. If $\Lambda = 0$, then $\alpha/(b\sqrt{d}) = 1$, i.e., $k = b\sqrt{d} = u^{n/2}$. But $k \in \mathbb{Z}$ and $u^{n/2} \notin \mathbb{Z}$ when n is odd and u is not a perfect square (since $u^{n/2} = u^m \sqrt{u}$ and $\sqrt{u} = a\sqrt{d} \notin \mathbb{Q}$). Hence $k \neq u^{n/2}$, so $\Lambda \neq 0$. $\square$

3.2.7 Combining the Bounds

From the upper bound (15):

$$\log |\Lambda| \leq \log 2 + \log |f(n)| - n \log u. \tag{22}$$

From the lower bound (21), substituting $B = O(n)$, $A_1 = \log \varepsilon$, $A_2 = \log u$, and $A_3 = O(\log |f(n)| + 1)$:

$$\log |\Lambda| > -C'_0 (1 + \log n) \log u (\log |f(n)| + C'_1), \quad (23)$$

where $C'_0 = C_0 \log \varepsilon$ and C'_1 are constants depending only on u (not on n or f).

Combining (??) and (??):

$$n \log u < \log |f(n)| + \log 2 + C'_0 (1 + \log n) \log u (\log |f(n)| + C'_1). \quad (24)$$

Theorem 3.8. *Under the growth condition (2), every solution of (3) with $f(n) \neq 0$ satisfies*

$$n \leq N_0(u, f),$$

for an explicit constant $N_0(u, f)$ depending only on u and the growth rate of f . In particular, the number of such solutions is finite.

Proof. We show that (24) cannot hold for arbitrarily large n . Dividing both sides by $n \log u$:

$$1 < \frac{\log |f(n)|}{n \log u} + \frac{\log 2}{n \log u} + \frac{C'_0 (1 + \log n) (\log |f(n)| + C'_1)}{n}.$$

By the growth condition (2), $\log |f(n)| = o(n/\log n)$. Hence:

- $\frac{\log |f(n)|}{n \log u} = o\left(\frac{1}{\log n \cdot \log u}\right) \rightarrow 0.$
- $\frac{(1 + \log n) \log |f(n)|}{n} = (1 + \log n) \cdot o\left(\frac{1}{\log n}\right) = o(1) \cdot \frac{1 + \log n}{\log n} \rightarrow 0.$
- $\frac{(1 + \log n) \cdot C'_1}{n} \rightarrow 0.$

So the right-hand side tends to 0 as $n \rightarrow \infty$, while the left-hand side is 1. This is a contradiction for n sufficiently large.

More precisely, there exists $N_0 = N_0(u, f)$ such that for all $n > N_0$, the right-hand side of (24) is less than $n \log u$, making the inequality impossible. This N_0 can be made explicit by tracking the constants; it depends on u (through $\log u$, $\log \varepsilon$, C'_0 , C'_1) and on the implicit constant in the $o(\cdot)$ of (2).

Since every non-trivial solution has $n \leq N_0$, and for each n there is at most one $k = \sqrt{u^n + f(n)}$ (when it is a perfect square), the total number of solutions is at most N_0 . $\square$

3.3 Assembling the Proof of Theorem 1.1

Proof of Theorem 1.1. We consider all positive-integer solutions (n, k) of (3).

Case u is a perfect square ($d = 1$): By Proposition 3.1, every solution with $f(n) \neq 0$ has $n \leq N_1$ for some explicit N_1 . Solutions with $f(n) = 0$ give $k = v^n$ (trivial); if there are infinitely many such n , they constitute a degenerate family, and the non-trivial solutions are finite.

Case u is not a perfect square ($d > 1$):

- Even n : By Proposition 3.2, non-trivial solutions have $n \leq N_2$.
- Odd n : By Theorem 3.8, non-trivial solutions have $n \leq N_0(u, f)$.

Taking $N = \max(N_1, N_2, N_0)$, every non-trivial solution satisfies $n \leq N$, and the total number of solutions is at most N (plus the trivial family if f vanishes infinitely often). In particular, the solution set is finite. $\square$

4 Discussion and Remarks

4.1 Sharpness of the Growth Condition

The growth condition (2), namely $\log |f(n)| = o(n/\log n)$, is essentially optimal for the Matveev-based approach. The bottleneck is the factor $(1 + \log n) \cdot A_3$ in the Matveev lower bound (21), where $A_3 = O(\log |f(n)|)$. For the upper bound (22) to contradict the lower bound, we need

$$n \log u > C(1 + \log n) \log |f(n)|,$$

which requires $\log |f(n)| = o(n/\log n)$.

If f grows faster—say $|f(n)| = \exp(\Theta(n/\log n))$ —the two bounds become comparable and the argument breaks down. Whether the finiteness conclusion still holds for such f is an interesting open question that would require sharper tools (e.g., the p -adic Baker–Wüstholz method, or the Laurent–Mignotte–Nesterenko refinement with better dependence on A_3).

4.2 Comparison with the Reference

The reference [1] treats the special case $u = 2$, $f(n) = -n$, where $\log |f(n)| = \log n$. In this case:

- The Pell equation is $k^2 - 2q^2 = -n$ with $q = 2^m$, working in $\mathbb{Q}(\sqrt{2})$ with unit $\eta = 3 + 2\sqrt{2}$.
- The linear form is $\Lambda = \log \gamma + t \log \eta - m \log 2$ with $\gamma = \beta/(2\sqrt{2})$, exactly matching (17).
- The Matveev bound gives $\log |\Lambda| > -2.31 \times 10^{12}(1 + \log m)(\log n + 3.843)$, and the upper bound gives $|\Lambda| < n/2^{n+1}$, yielding $n \leq 4.85 \times 10^{15}$.

Our generalization replaces the specific constants $(\eta, \log 2, 3.843)$ with their analogs $(\varepsilon, \log u, O(1))$ and replaces $\log n$ (from $f(n) = -n$) with the general $\log |f(n)|$, but the structure of the proof is identical.

4.3 Effective Computability

The bound $N_0(u, f)$ is in principle *effective*: given u and an explicit f (with a known growth rate), one can compute:

1. The squarefree decomposition $u = da^2$ and the fundamental unit ε of $\mathbb{Q}(\sqrt{d})$.
2. The Matveev constant C_0 (a universal constant $\approx 2.31 \times 10^{12}$).
3. The threshold N_0 beyond which $\log |f(n)| < n \log u / (2C_0(1 + \log n))$.

For specific (u, f) , one can then (in principle) perform the quadratic-residue sieve of [1], Section 5, to reduce the candidates to a finite list and check each directly. The sieve method generalizes verbatim: the condition “ $u^n + f(n) \bmod p \in \text{QR}(p)$ ” depends on n modulo $\text{lcm}(p, \text{ord}_p(u))$, and the “wheel” construction of Algorithm 2 in [1] applies with u in place of 2.

4.4 The Case $u = 1$

As noted in Remark 1.3, the theorem fails for $u = 1$: the equation $1 + f(n) = k^2$ can have infinitely many solutions (e.g., $f(n) = n^2 - 1$). The reason is that $u^n = 1$ does not grow, so the “perturbation” $f(n)$ is not small relative to u^n , and the linear form Λ is not forced to be exponentially small. The proof breaks down at Lemma 3.5: $|\Delta| = |f(n)|/u^n = |f(n)|$, which is not small.

4.5 Extensions

The same proof strategy applies, *mutatis mutandis*, to:

- Equations of the form $u^n + f(n) = k^r$ for fixed $r \geq 2$, using the Thue equation approach or higher-degree number fields.
- Equations $u^n - f(n) = k^2$ (the sign of f is irrelevant to the argument).
- Multi-term equations $u_1^{n_1} + u_2^{n_2} + f(n_1, n_2) = k^2$ in two variables, using S -unit equations and the Schmidt Subspace Theorem, though the analysis becomes significantly more involved.

References

- [1] Anonymous, *The equation $2^n - n = k^2$ has only the solutions $n = 1, 7$* , Preprint, September 2026.
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- [2] E. M. Matveev, *An explicit lower bound for a homogeneous rational linear form in logarithms of algebraic numbers. II*, *Izvestiya: Mathematics* **64** (2000), 1217–1269.
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- [4] A. Baker, *Transcendental Number Theory*, Cambridge University Press, 1975.
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